

Entrepreneurship Mid-Term Paper

Bidet Patent

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Introduction:

Many portable bidets are in the market today. The design proposed, however, has two key differences from those on the market. Hygiene is one major factor that comes with the bidets, the issue that shows up in most, if not all the designs on the market now is that feces can drop into the tube. The given design fixes this concern by implementing a removable tube into the design. The other difference is that the invention allows for both battery operated cleaning propulsion, and to be able to squeeze the bidet to expel the liquid.

For this invention, Stacy Consulting was brought to investigate possible patent conflictions, help find strategic partners, do market research, and help price the product.

Patent Research:

Stacy Consulting uses Google Patent to conduct our patent research.

Since one of the major applications of the invention is the disposable supply tube, this was the first patent search conducted. The first disposable tube that was found was patented by James Kenney. This is for a disposable capillary tube which uses a piston to suck fluids into the system, then the piston is used to eject the fluids. By not using a piston this patent will not block the supplied invention [1]. This was the only patent which had possibility of blocking the invention from the supply tube.

Next, looking at the other portable bidets on the market. One issue that arises is that the definition of portable seems to have changed over the years. Many old designs work with the definition of portable being that one can install the device onto any toilet. The invention in question works with portability being more well defined as can be brought from place to place, for out of the home use, no installation needed. One patent that was flagged was by Jacob and Aviv Bader, this one had a charging point connection on top of their device, part 54 in their diagram [2]. This could reflect the same as ours and cause the block from patenting. Suggestion would be to assure rights to such charging port from the original charging such as the patent created by Noriyuki Matsuda in 2001 [3].

In 2006, Jeffrey B Fogel patented a truly portable bidet, meaning no inline connection to the waterline. This one has the storage capability, which could cause an issue at the patent office. The difference is that your invention can completely disassemble and all fit inside of the main tube, where for Fogel's design, the tube is unscrewed, flipped around and screwed in the opposite way. Since yours completely disassemble this is considered different from that of Fogel's design [4]. Having the squeeze bottle, however, could be a significant issue, although the squeeze bottle itself is not patent protected here, the idea of this being a main way to project the fluid from the vessel could be considered. To avoid blockage of the patent, suggestion would be to get approval from inventor Frank M Harding on his squeeze bottle design [5].

-5
not full

Strategic Partnership:

When thinking of the strategic partners for this product, Stacy consulting immediately thought to look at the largest toilet companies. As for our search, the 8 largest companies were shown:

1. American Standard
2. Toto
3. Kohler
4. Duravit
5. Niagara Conservation
6. Sterling
7. Jacuzzi
8. Saniflo



These companies are listed in order, with the largest being at the top [6]. Many factors went into play for each company.

1. American Standard:

- Although the largest company American standard does not offer a truly portable bidet. They have a built-in bidet option, so they are willing to work into the bidet market. Being the biggest company, it may be hard to work with them, however, they also have more resources to spread around.

Suggestion- Attempt to Partner

2. Toto:

Toto, although the second biggest company would not be a good company to partner with. They already have their own truly portable bidet. Will be discussed more in the market share section listed below.

Suggestion - Do Not Partner

3. Kohler:

Kohler also falls under the category of having skin in the bidet market, just not a portable one. They have a built-in system, so they have shown interest in the market.

Suggestion- Attempt to Partner

4. Duravit:

Duravit has no portable bidet, but does have built in bidets that they sell, interest has been shown in the bidet market.

Suggestion – Attempt to Partner

5. Niagra Conservation:

Biggest company without any bidets. Placing 5th in size is no small feat, however, if they could work into the bidet market they could pass the companies above them. They are large enough that they could want to make the investment, but have not shown any interest in the market. Only want to attempt to convince one company without a bidet, to possibly want to create a portable one.

Suggestion – Attempt to Partner

6. Sterling:

Since Sterling does not have a bidet, no interest has been shown.

Suggestion – Do Not Partner

7. Jacuzzi:

Also has shown no interest in the bidet market.

Suggestion – Do Not Partner

8. Saniflo:

With Saniflo being the smallest company in our list, and no interest in the market, I would suggestion to not partner with them.

Suggestion – Do Not Partner

Market Share:

According to mental floss 80% of the middle east and japan use bidets [7]. Promodeler confirmed the 80% number [8]. With a quick google search I was able to obtain that there are roughly 411 million people in the middle east [9], and roughly 127 million people in japan [10]. Adding these number together meant that the total population of these places yield 538 million. Taking 80% of this number means that roughly 430 million people use bidets.

Obviously saying that everyone that uses a bidet at home would want to own a portable bidet is an outrageous claim. Without being able to survey anyone who uses bidets, and educated guess based off my companies' experience was used. Assuming roughly 50% of people who use bidets would want to use a bidet outside of the home. Although I would say that around 30% of that market already has their own portable bidet that they like to use, and the other 20% doesn't have enough interest in purchasing one. Since your product has a higher hygiene aspect, I believe that at least 10% of the customers who already own a portable bidet would shift to using your product. Then I believe that with all the upgrades to your product compared to other inventions, you would be able to pull around 5% of the people who have not purchased one to go out and try it. Especially since the low initial cost doesn't break the bank for an electric bidet, cost explained later in the competitive pricing portion. Leading to roughly 35 million potential customers.

Competitive Pricing:

As previously stated, many companies are in the truly portable bidet market today. However, your invention allows them in have a much smaller item, with more functionality. Other companies include:

- Sonny - \$99 with a discount (original \$140) [11]
Battery has a 3-week charge, with a 2 separate spray setting. Their product is only in the beginning stages, with only being able to sell limited stock. It as a little over 9,000 backers that support it.
Your invention – Better because of the dual threat of being able to hand pump if needed. And the replaceable stem allows for your product to not need to be replaced.
- Tushy - \$29 [12]
Very small, so well portable, however, it doesn't include any electrical flow push. Also has no was of swapping out the nozzle, so if it gets dirty it would need to be replaced.
Your invention – Better because of the dual threat of being able to hand pump if needed. And the replaceable stem allows for your product to not need to be replaced.

- Bio Bidet - \$10.24 [13]

Large device which comes with carrying case. Controls water flow by airlock system, however, only flow is from squeezing the bottle. No electronic flow. Material is cheap and looks more to be a disposable bidet after a few uses since it comes in multiple pack sizes. Company is based in Illinois, so could be competitor close to home if chosen to create product in Illinois.

Your invention – Blows this away, no comparison to be made.

- Toto - \$83.45 [14]

Most well known company. The bidet is small to be able to be brought everywhere, then at the click of a button releases into a larger apparatus which allows for better use. Has 2 settings for water flow that would be allowed out of it. Has no ability to replace the head of the bidet.

Your invention- Will struggle to compete with this large of a company. However, yours allows for the dual water flow, and has the cleanliness factor going for it too.

Without knowledge of how much the product costs to make and based off the competitors pricing I believe we I would suggest sell your product for around \$40. I know this seems to be to small of a number. However, I believe that your invention must take the route that razers and printers have, the device itself is not all that expensive, however, you sell replacement tubes. A 12 pack of tubes could cost up to \$20.

References:

Patents:

- [1] Kenney, J. W. (n.d.).
- [2] Bader, J., & Bader, A. (n.d.).
- [3] Matsuda, N. (n.d.).
- [4] Fogel, J. (n.d.).
- [5] Harding, F. M. (n.d.).

Strategic Partners:

- [6] "Best Toilet Brands in the Market You Should Trust." *ToiletsMan*, 29 Nov. 2018

Market Share:

- [7] "Why Aren't Bidets Common in the U.S.?" *Mental Floss*, 28 June 2013
- [8] "Bidets Finally Making Inroads in US Bathrooms." *Pro Remodeler*, 25 Sept. 2017
- [9] "The Middle East Population 2019." *The Middle East Population 2019 (Demographics, Maps, Graphs)*
- [10] "Japan Population (LIVE)." *Worldometers*

Competitive Pricing:

- [11] Sonny. "Sonny: A Portable Bidet for Eco-Conscious Humans." *Indiegogo*, 9 July 2019
- [12] "TUSHY: TUSHY Travel." *Portable Bidet*
- [13] Schwartz, Alvin, and Samuel Nocella. "Stores." *Amazon*, Macmillan, 1977.
- [14] "Travel Washlet®." *TotoUSA.com*